

AUTO-ANNOTATOR

Full-Stack Enterprise MLOps Command Center

Overview: A full-stack, enterprise-grade AI platform that utilizes an autonomous agent swarm to generate, self-verify, and label training data, before training deployable YOLO object detection models natively on AMD hardware. Features a robust UI dashboard while remaining fully API-extensible.

1. THE PROBLEM & MARKET GAP

The "Cold Start" problem in computer vision is massive. Enterprises (manufacturing, drones, medical) need custom object detection, but face two massive bottlenecks:

- **Manual Labeling Tax:** Paying humans to draw bounding boxes on 10,000 images is incredibly slow and expensive.
- **Data Scarcity & Privacy:** Scraping web images carries heavy legal/copyright risks. Sending proprietary factory floor images to public APIs violates internal data privacy policies.

2. COMPETITOR ANALYSIS

"Doesn't Ultralytics or Roboflow already do this?" No. They are isolated tools, not autonomous agents.

Feature	Roboflow / Ultralytics	Our Agentic Swarm
Autonomy	Human must click, upload, review, and trigger training.	Zero-human intervention. Give it a prompt, get a .pt model.
Data Sourcing	User must bring a pre-existing curated dataset.	Can autonomously generate "synthetic" data from scratch.
Verification	Model hallucinations ruin the dataset.	Critic Agents self-correct geometric bounding-box errors.
Infrastructure	SaaS Cloud (Privacy Risk).	100% Local Enterprise deployment optimized for ROCm.

3. THE ENTERPRISE COMMAND CENTER (DASHBOARD INTERFACE)

To provide immediate commercial viability, the system features a comprehensive graphical dashboard, transitioning the project from a headless API to a complete **MLOps Control Plane**. This allows non-technical operators to manage the entire AI lifecycle:

- **Live Agent Observability:** Real-time terminal streaming of agent reasoning, Critic Agent rejection logs, and pipeline state transitions.

- **Synthetic Data Foundry UI:** Visual controls to adjust domain randomization parameters, preview generated SDXL/Flux batches, and curate datasets before labeling.
- **Model Registry & Analytics:** Centralized management of trained YOLO weights (.pt), showcasing validation metrics (mAP, Precision, Recall) and dataset distribution statistics.
- **Hardware Telemetry:** Live tracking of AMD MI300X VRAM utilization, GPU compute spikes during training phases, and inference throughput.

4. THE DUAL-PATH ARCHITECTURE

Path A: The Synthetic Foundry (For the "Cold Start")

User inputs a prompt via the UI. A **Prompt Agent** expands this into 500 scenarios using "Domain Randomization". A **Synthesis Agent** runs SDXL/Flux to generate images royalty-free.

Path B: BYOD Enterprise (Bring Your Own Data)

A business uploads a secure `.zip` file of their own unlabelled proprietary factory images via the Dashboard.

The Core Engine (Convergence)

1. **Vision Agent:** Loads SAM 3 (Segment Anything) to generate zero-shot pixel masks of the targets.
2. **Critic Agent:** Uses OpenCV to verify geometry, convert masks to tight YOLO bounding boxes, and format the dataset.
3. **MLOps Agent:** Flushes the VRAM, compiles the dataset, and triggers a PyTorch YOLOv10 training loop natively via AMD ROCm.

5. WHY THIS WINS THE AMD HACKATHON

- **Max Hardware Flex:** The pipeline purposefully runs three heavy sequential workloads (Generation -> SAM 3 -> Training). This proves the power of the MI300X's 192GB VRAM.
- **VRAM Orchestration:** The system intelligently flushes GPU memory (`hip.empty_cache()`) between pipeline stages, showcasing deep hardware awareness.

6. TECHNICAL STACK DETAILS

- **Hardware:** AMD Instinct MI300X (via Developer Cloud)
- **Frontend Dashboard:** Streamlit or Next.js/React
- **Vision & Generation:** SAM 3, SDXL/Flux (HuggingFace diffusers)
- **Logic:** LangChain/CrewAI, OpenCV, FastAPI
- **Optimization:** AMD ROCm, PyTorch Inductor